OFFICIAL MODEL SPECIFICATION

Nexus-70B-Instruct

Version: v1.2.0 • **License:** Apache-2.0 with Responsible AI Deployment Addendum • **Type:** Dense Decoder-only Transformer LLM • **Released:** 2026-08-15

Nexus-70B-Instruct is a high-capability generative language model optimized for complex reasoning, multi-turn coding, and instruction following across enterprise workflows.

BENCHMARKS	PEAK SCORE	MAX DELTA	EVAL RUN ID
3	**93.6%**	**+6.7%**	**eval-run-202**
Multi-domain	Standardized	vs Nexus-70B-v1	2026-08-17

1. Architecture & Governance

Model Specifications

Parameters: 70 Billion parameters
Layers: 80
Heads: 64
Kv Heads: 8
Context Window: 131,072 tokens
Activation: SwiGLU
Positional Embeddings: RoPE (Rotary Position Embeddings) with theta=500000

Governance & Ownership

Organization / Lab: Nexus AI Research Foundation
Maintenance Contact: Nexus Foundation AI Safety & Alignment Group — ai-safety@nexus-models.example.org
Hardware & Compute: *Not provided*

2. Operational Scope & Limitations

Primary Intended Applications

• Complex reasoning, quantitative analysis, and mathematical problem solving
• Software engineering assistance and multi-language code generation
• Technical document summarization and executive brief preparation
• Enterprise agentic tool use and workflow orchestration

Known Limitations

• Context window limited to 128k tokens; attention quality degrades on >100k distractors
• May occasionally hallucinate citations when querying real-time world events beyond knowledge cutoff
• Non-English low-resource language performance is lower than high-resource benchmarks

Prohibited & Out-of-Scope Deployments

• Autonomous medical diagnosis or clinical prescription without certified human-in-the-loop oversight
• Generation of malicious cyber-attack scripts, malware, or exploit payloads
• High-frequency algorithmic trading decision making without backtesting guardrails
• Automated legal sentencing, credit denial, or biometric surveillance decisions

3. Quantitative Evaluation Benchmarks (vs Nexus-70B-v1.1)

Last evaluation cycle: 2026-08-17 10:30:00 UTC • Run ID: eval-run-20260817-v1.2

Benchmark	Domain	Metric / Setting	Score	Baseline (Nexus-70B-)	Delta	Details
MMLU (Massive Multitask Language Understanding)	General Knowledge & Reasoning	Accuracy (5-shot) (5-shot)	**88.4%**	84.1%	▲ **+4.3%**	57 subjects across STEM, humanities, social sciences
GSM8K (Grade School Math 8K)	Mathematical Reasoning	Exact Match (Pass@1) (8-shot)	**93.6%**	89.2%	▲ **+4.4%**	Multi-step grade school math word problems
MATH	Advanced Mathematics	Exact Match (4-shot)	**68.2%**	61.5%	▲ **+6.7%**	Challenging competition math problems

4. Training Data & Safety Evaluations

Training Data Summary

Total Tokens: 15.2 Trillion tokens
Composition: Curated web crawl (55%), Technical papers & textbooks (20%), Synthetic reasoning datasets (15%), High-quality source code (10%)
Cutoff Date: March 2026
Filtering: Strict toxicity, PII redaction, deduplication, and synthetic quality scoring

Safety Alignment & Ethics

• Extensively red-teamed against prompt injection, jailbreaking, and hazardous chemical/biological requests
• Reinforcement learning with Human & AI Feedback (RLHF/RLAIF) applied for helpfulness and safety
• Carbon emissions offset verified: 120 tCO2eq neutralized via certified carbon credits

5. Citation & Reference

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@article{nexus2026,
title={Nexus-70B: Technical Report on Frontier Reasoning and Agentic Capabilities},
author={Nexus Research Team},
journal={arXiv preprint arXiv:2608.01234},
year={2026}
}
```